RMSE reduction relative to paired GraphCast · di16 fast-Mamba SWA 500–2000

2 m temperature

Mean sea-level pressure

10 m zonal wind (u)

10 m meridional wind (v)

6 h precipitation

Geopotential · all levels

Air temperature · all levels

Specific humidity · all levels

Zonal wind (u) · all levels

Meridional wind (v) · all levels

Pressure vertical velocity · all levels

RMSE reduction (%) · positive means lower error

Forecast lead (days)

2022 development · 32 starts

2023 pilot · 8 starts